Zero45 – Background for the game master

Model, data, design decisions and answers to typical questions · as of 18 October 2026

Game idea based on “Path to Zero” by Jesse D. Jenkins and the Princeton ZERO Lab · Zero45 is an independent adaptation · Free software (GPL-2.0-or-later) · Learning game, not a forecast

> **In one sentence:** Teams build a German electricity system in four stages (up to 2030, 2035, 2040, 2045); each stage is calculated with a real electricity market model (PyPSA, solver HiGHS) for all 8,760 hours of a year and scored on security of supply, climate protection (CO_2-free share *and* CO_2 emissions) and costs. Guiding principle: **the numbers come from official sources (TYNDP 2024, Federal Network Agency BNetzA, Grid Development Plan NEP, PyPSA technology-data); wherever the game simplifies, this is documented.**

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1. Game flow and points

Each stage: **Build** (spend build tokens, keep/retire old power plants, use influence tokens) → **Test** in the simulator (3× per stage, normal year, no chance; from the second test with a comparison to the previous one) → **Mark as ready** (can be changed until the game master closes the phase; notice at 2 minutes left) → **Simulate the year** with shared chance for all teams → **Points** with a 💡 tip (what cost the most points, what changes in the next stage). At the start of each stage the projector shows in large print what is new. Afterwards learning effects and protests per team.

Goal	Measure	Rationale
Security of supply	Hours with a supply gap (> 100 MW missing)	German reliability standard: 2.77 h/year (BNetzA).
Climate protection	**The worse** of two values: (1) CO_2-free share = (demand + exports − imports − fossil generation) ÷ demand, max. 100 %; (2) CO_2 emissions in Germany in Mt/year	(1) Official Renewable Energy Act (EEG) logic: exported green power counts, imports do not – legal target 80 % in 2030. (2) Net-zero logic of the Climate Protection Act: tonnes, territorial, exports offset nothing. Together they close both loopholes (export credit, import relocation). **5 points = political target reached.**
Affordability	System costs per MWh of demand (excl. grids, taxes, levies)	Calibrated with bots; good teams can beat the official path (66–78 €/MWh).

Thresholds 5 / 4 / 3 / 2 / 1 points	Gap at most	CO_2-free at least	CO_2 at most	Costs at most
Stage 1 (2030)	3 / 10 / 25 / 50 / 100 h	80 / 75 / 70 / 65 / 60 %	70 / 80 / 90 / 105 / 120 Mt	58 / 63 / 68 / 73 / 79 €
Stage 2 (2035)	same	88 / 84 / 80 / 76 / 72 %	45 / 55 / 65 / 80 / 95 Mt	64 / 69 / 74 / 80 / 86 €
Stage 3 (2040)	same	93 / 90 / 87 / 84 / 80 %	25 / 35 / 50 / 65 / 80 Mt	63 / 68 / 74 / 80 / 87 €
Stage 4 (2045)	same	97 / 95 / 92 / 89 / 85 %	8 / 20 / 35 / 50 / 65 Mt	65 / 70 / 76 / 83 / 91 €

Anchors of the CO_2 thresholds (5 points): 2030 Climate Protection Act, annex 2a – energy sector 108 Mt (incl. district heating, refineries; electricity roughly 70 Mt, own estimate); 2035 G7 pledge Elmau 2022 “fully or predominantly decarbonized power sector” (no number – set in the game so that it is reachable with the game's means); 2040 coal exit 2038 and end of new allowances in the EU emissions trading system around 2039/40 (reform ongoing); 2045 net greenhouse-gas neutrality (waste incineration alone ~5 Mt). The share thresholds after 2030 rise to 97 % in 2045: the legal target is then climate neutrality (tonnes), and the EEG names a fully renewable power supply in the long run. The share rewards domestic green power over power imports and prevents a team from “outsourcing” its CO_2 via imports. Deliberately not 100 %: otherwise a narrow export surplus would decide the points. The 5-point CO_2 targets are the legal anchors; the lower levels are deliberately wide (half the way also earns points), and the cost thresholds are set so that simple strategies also earn points – otherwise the teams hardly differ from 2040 on and it becomes frustrating. Calibration: the net-zero bots reach 98.5–100 % in 2045 (thanks to exports), a strongly import-dependent net-zero path (≈ 150 TWh imports) 89 % (2 points).

Max. 15 points per stage, 60 per game. Tie: less CO_2 wins. Last stage: −2 points per protest against a new technology (otherwise everything would be built without a second thought at the end).

2. Tokens, influence, chance

Element	Value	Rationale
Build tokens	1 token = €10 bn; 18 / 18 / 17 / 17 per stage; unused ones expire	More at the beginning, like the real need: EEG expansion path wind+PV ≈ 204 / 157 / 120 / 74 bn € (total ≈ 555 bn; own estimate from the EEG expansion targets and technology costs). Early plants work in all stages – front-loading gave ~6 points more in bot tests than the same tokens spent late.
Credit tokens	up to 4 per stage, +20 % costs	Financing surcharge for “more now”.
Influence tokens	2 at the start; more for 2 build tokens each (counts as programme costs, annuity over 20 years); each measure once	As in the original game (1 budget token = 2 build tokens).

Measure	Effect	from
Climate resilience	impact of disasters halved	this stage
Accelerated innovation	double learning rate of your own construction projects	this stage
Hydrogen demonstration programme	own green hydrogen: unlocks electrolysis, H_2 storage and H_2 conversion (electrolysis only with this card)	next stage
Hydrogen grid & imports	pipeline hydrogen for converted gas turbines: 2035: 15 GW at 140 €/MWh H_2, 2040: 30 GW at 125 €/MWh H_2, 2045: 45 GW at 110 €/MWh H_2 (≈ 6 / 12 / 18 GW electricity); unlocks H_2 storage and conversion, no electrolysis	next stage (from 2035)
Acceptance & participation	protest risk halved	this stage
European grid expansion	+10 GW interconnectors	next stage
Flexible demand	load shifting 5/10/15/20 GW (within a day) + demand reduction 4/6/8/10 GW (300 €/MWh, max. 200 full-load hours)	next stage

Chance (shared by all teams in a stage)

- **Weather year:** normal 2008 (60 %), difficult 1995 (30 %, less wind, largest 2-week deficit), extreme winter 2009 (10 %, highest peak and residual load; roughly every 10–20 years). The simulator always calculates 2008.
- **Demand:** ±3 % standard deviation (capped at ±10 %).
- **Disasters:** probability 20 / 30 / 40 / 50 % per stage; storm/flood 40 % (each technology hit separately with 40 % → half output for 4 weeks), cold dark doldrums 35 % (2 weeks Nov–Feb: wind 30 %, PV 50 %,

demand +5 %), heat & drought 25 % (4 weeks Jun–Sep: gas/coal/biomass/oil 80 %, run-of-river 60 %, demand +3 %). Pumped hydro, waste and imports are not affected.

Team-specific

- **Protest (backlash):** new technologies per token none/low/medium/high = 0/2/5/10 %; kept old power plants once per stage 0/5/10/25 %. New technology: what was built keeps running, one stage building ban. Old power plant: retired completely and permanently in the next stage. Risks: onshore wind, ground-mounted PV, new gas turbines medium; coal high; rooftop PV, batteries, run-of-river none.
- **Learning effects:** per token built on average −1 % costs (random, spread ±0.5 %), double with innovation – on top of the cost trend in the data.

3. The model

Type: linear optimisation model (PyPSA 1.3, solver HiGHS) for Germany as **one grid node**, 8,760 hours, **perfect foresight** over the year, cyclic storage (end of year = start of year). The teams set the capacities; the model only determines the **least-cost dispatch** hour by hour.

- **Objective:** minimal operating costs (fuel, CO_2, variable costs, imports – exports) plus unserved energy at 10,000 €/MWh (value of lost load).
- **Supply gaps:** the model minimises the missing energy. A tie-break (gaps in high-load hours 1 % "cheaper") puts remaining gaps into the peak hours so that the number of hours is unambiguous. An hour counts from 100 MW missing. Because of perfect foresight and no power plant outages, the number is rather a **lower bound**.
- **Demand:** net consumption excl. electrolysis 560/760/875/925 TWh, plus grid losses and power plant own consumption (+7/6.5/6/6 %) → **≈ 600 / 810 / 930 / 980 TWh**; peak load in a normal year 106 / 143 / 163 / 166 GW (about +20 % in the extreme winter 2009).
- **Computing time:** 10–20 s per team and year; up to ~1.2 GB of memory per run; in parallel according to cores and free memory.

4. Technologies and existing fleet

New builds (costs 2030, falling until 2045)

Technology	Investment	per token	Capacity factor	Assumptions
Onshore wind	1,383 €/kW	~7.2 GW	26–29 %	30 years; PECD profile of the stage
Offshore wind	2,749 €/kW	~3.6 GW	~45 %	incl. ~30 % grid connection; ×0.85 for wake losses in wind farm clusters (IWES)
Ground-mounted PV	482 €/kW	~20.7 GW	~11 %	PECD profile ×1.10 (orientation, Fraunhofer ISE), 40 years
Rooftop PV	884 €/kW	~11.3 GW	~10 %	no protest risk
Battery storage	973 €/kW	~10.3 GW	–	energy content 4 h at full power, 92 % efficiency (charging+discharging), 20 years
Electrolysis	1,886 €/kW	~5.3 GW	by surplus	62–68 % efficiency, 25 years
H_2 storage (salt cavern)	2.67 €/kWh	~3.7 TWh H_2	–	100 years; injection/withdrawal capacity not limited
New gas turbines (H_2-ready)	640 €/kW	~15.6 GW	dispatchable	OCGT 41 %; conversion to H_2 +120 €/kW
Biomass flexibilisation	1,201 €/kW	~8.3 GW	dispatchable	more capacity for the same energy (41 TWh), max. 3× the biomass kept

Existing fleet (BNetzA power plant list, 26 June 2026)

- **Keep or retire (permanently):** coal 21.9 GW (legal cap 17 / 6 / 0 / 0 GW; fleet block: min. 15 % output, ramp 40 %/h), gas 30.1 GW (50 %), biomass 8.9 GW (41 TWh/year). Run-of-river 3.9 GW always runs (listed with the fixed plants).
- **Old wind and PV plants age out:** onshore wind 69.9 GW → 82/68/49/30 % (weaker profile, Ø ~20 %), offshore 10.6 GW (Ø ~34 %), PV 110.6 GW → 100/98/84/65 %.
- **Fixed plants with a schedule:** pumped hydro 9.9 GW (6 h, 77 % round trip), old large batteries 3.7 GW (1 h, gone by 2045), waste 1.9 GW (always running, half renewable), oil & other 4.8 GW (gone from 2040), reserve 11.1 → 3.6 GW (500 €/MWh, only in emergencies), interconnectors 25/28/32/35 GW.

5. Neighbouring countries, flexibility, hydrogen

Neighbouring market (imports/exports)

- The 12 neighbouring bidding zones (AT, BE, CH, CZ, DK-E/W, FR, LU, NL, NO-South, PL, SE4) are **one synthetic market**: TYNDP demand, their own PECD weather (same weather year → correlated dark doldrums), wind/PV shares according to the EU mix TYNDP NT+, nuclear/hydro/bio as a constant block. From this an hourly “fossil need” F(t).
- **Imports:** available is the neighbours' spare power plant capacity (1.05 × max F − F(t)); price and CO_2 rise with scarcity: surplus 10 €/MWh CO_2-free → CCGT costs → gas turbine → 300 €/MWh; +12 €/MWh fee.
- **Exports:** the neighbours buy what replaces their fossil plants (market price − 5 €), plus at any time up to 8 GW of surplus at 2 €/MWh (saving hydropower, electrolysis, heat).
- **Storage abroad:** 0.5 TWh, 85 % per direction: takes German surplus, returns it later at the import price; shares the interconnector.
- **Calibration:** the 2030 start fleet imports ~75 TWh (2024: 67 TWh); the TYNDP fleet 2030 exports 49 TWh (TYNDP: 54).

Flexible demand (only via influence token)

Load shifting = lossless storage with energy for 4 h that has to balance out **within each day** (e.g. evening → noon). Demand reduction = industry cuts back from 300 €/MWh, max. 200 full-load hours. Values deliberately below NEP/TYNDP (there 14–28 GW explicitly plus flexible households).

Hydrogen

Three building blocks that can be **expanded separately**: **electrolysis** (GW; runs only on the German green-power surplus of the hour – green hydrogen, following the EU rules on temporal correlation) → **salt cavern** (TWh H_2) → **converted gas turbines** (41 %, afterwards hydrogen only, no natural gas). Round trip ~25–28 %. Conversion is only possible when electrolysis and storage are built or planned; the interface shows the range (“enough for X days at full load”) and warns when a building block is missing. The storage is built separately because a fixed coupling to the electrolysis would be far too small for a long-term reserve. In addition there is the influence card **“Hydrogen grid & imports”** (pipeline H_2 at 140 → 110 €/MWh H_2, 15 / 30 / 45 GW from 2035): without it, climate neutrality is practically out of reach in the game – the calibration showed this (see section 8); with it, it is possible, but expensive. The two hydrogen cards have separate roles: demonstration programme = electrolysis, storage, conversion; grid = storage, conversion, imports. Own electrolysis plus imports costs two tokens.

Firm capacity (display)

“Firm” only counts dispatchable power plants (gas, H_2 turbines, coal, biomass, oil, reserve; run-of-river half, waste partly). Storage & flexible demand and interconnectors are listed below as separate lines. Three buffers: “power plants only” (grey, strictly by definition, applies to lulls lasting several days), “with storage & flex” (traffic light: green from +10 %, yellow from 0 %, covers the evening peak) and “with storage, flex & interconnectors” (stricter traffic light: green from +30 %, yellow from +15 %, because imports are missing in a Europe-wide lull). Checked with test runs for 2040 (portfolio with lots of storage, gas varied): “+ storage & flex” +5 % → difficult year 0 gap hours, extreme winter 6; +15 % → 0 and 1. Interconnectors add about 20 percentage points. The GW values are behind the info button. Gas turbines converted to hydrogen only count as far as imports and a full

storage can supply them for two weeks (convention: the length of the cold dark doldrums in the game; in the model, electrolysis can refill). Display only – scoring is hourly in the model.

6. Costs and CO_2

- **Costs per stage** = annuities of all investments so far (5 % real, over the lifetime; count in all later stages) + fixed operating costs + fuel, CO_2, variable costs + imports − exports + credit surcharge + influence programmes, divided by demand. Existing plants only with operating costs (their capital is "sunk"). Missing power is shown as damage (10,000 €/MWh) but does not count for the cost points – otherwise a gap would cost twice (supply 0 and costs 0); in tests with human-like strategies a single difficult stage dropped from 10–12 to 3 points. It cannot be exploited: whoever does without firm capacity only saves 1–2 €/MWh but loses up to 5 supply points.
- Prices in 2025 euros (no inflation). Fuel/CO_2 according to TYNDP 2024 (IEA WEO 2022 APS): CO_2 134 / 154 / 173 / 186 €/t; natural gas 26.7 → 22.7 €/MWh_{th}.
- **CO_2 emissions** only in Germany (territorial as in the Climate Protection Act); factors TYNDP (gas 0.205, hard coal 0.338, lignite 0.364 t/MWh_{th}); waste 0.5 t/MWh_{el}. The CO_2 of imports is shown but does not count.
- **CO_2 per kWh** (display): emissions ÷ demand as with the German Environment Agency UBA (2024: 353 g/kWh, preliminary; 2025 estimated 344 g/kWh; UBA 2026) – imports with zero; additionally a second value "incl. import CO_2".
- **Two logics for imports:** for the share, imports count as non-renewable (EEG statistics), for emissions as emission-free (territorial). Net zero (Climate Protection Act) counts tonnes in Germany; exports offset nothing.

7. Data sources

Area	Source	Processing
Wind/PV hourly	ENTSO-E PECD 3.1 (TYNDP 2024), DE00, climate years 1995/2008/2009, target years 2030/2040	Stages 2030/35 use 2030, 2040/45 use 2040; corrections see section 4
Demand hourly	TYNDP 2024 demand profiles (NT 2030, NT 2040, Distributed Energy 2050)	Shape only; 2035 = mean of NT30/NT40, 2045 = mean of NT40/DE50; scaled to the annual values (normal year), other weather years keep their ratio
Demand level	Energy transition monitoring report 2025 (2030), NEP 2037/2045 (2025) middle path	Net excl. electrolysis + grid losses/own consumption
Run-of-river	TYNDP 2024 hydro inflows (PEMMDB DE00)	Profile × capacity
Fuel, CO_2	TYNDP 2024 commodity prices; CO_2 factors v2	EUR 2021 → 2025 (×1.18 Destatis CPI)
Technology costs	PyPSA technology-data (mainly Danish Energy Agency), EUR 2025	per stage year
Existing fleet	BNetzA power plant list 26 June 2026; PV split rooftop/ground-mounted 66/34 % (MaStR)	Ageing by commissioning cohorts (assumption)
Neighbouring countries	TYNDP demand + PECD of the 12 neighbours, EU mix NT+ (scenario report)	Synthetic market, calibrated to TYNDP trade volumes
Coal exit	Coal Phase-out Act (KVBG)	max. 17 / 6 / 0 / 0 GW
Hydrogen import price	Own assumption within the ranges of EWI 2025 ("Global production and delivery costs of green hydrogen": pipeline imports 2030 ~150 €/MWh) and Fraunhofer HYPAT 2024 (2030: 3.5–6.5 €/kg, 2050: 2–4.5 €/kg)	140 / 125 / 110 €/MWh H_2 (2035/40/45) ≈ €4.2 / 3.8 / 3.3 per kg

All values with sources in `ptz/config/assumptions.yaml`, game parameters in `game_default.yaml`. Processed data (~1 MB) are included; raw data (1.4 GB) only needed to regenerate them.

8. Calibration and comparison with studies

Game balance

- Adaptive bot (tries up to 10 plans per stage in the simulator, incl. partial conversion, hydrogen imports and retiring old gas plants), 7 runs with the card “Hydrogen grid & imports”: **48–55 of 60 points**. In 6 of 7 runs **net zero** is reached by 2045 at the latest (5–7 Mt), once already in 2040; without hydrogen it stays at ~40 Mt. Costs are then 67–71 €/MWh. **Risk:** in 3 runs the bot retired all old gas plants and had 23–43 gap hours in the difficult or extreme weather year (costs up to 107 €/MWh) – climate-neutral without a reserve is risky, just like in reality.
- Typical CO_2 path of good teams: ~45–50 Mt (2030) → 52–56 (2035, big demand jump) → 39–50 (2040) → 5–40 (2045). Whoever does not plan for hydrogen gets only 0–3 climate points in 2040/45 instead of 5.
- **Key insight:** with own electrolysis alone, emissions stay at ~45–55 Mt (old gas plants have to cover the dark doldrums, own hydrogen is not enough). In the game – as in the official scenarios – climate neutrality needs hydrogen from abroad. Hence the card “Hydrogen grid & imports”.
- Fixed, “naive” strategies score clearly lower (supply gaps or high emissions).

Comparison with official scenarios (the study's power plant fleet calculated in the game model)

- **Yields:** PV ±1 %, wind in total −2 % compared with the NEP; onshore wind ~10 % lower, offshore between NEP (3,330 h) and TYNDP (4,300 h).
- **2030:** the EEG target fleet (115/30/215 GW) reaches ~79 % CO_2-free at ~650 TWh demand – fits the 80 % logic. 0 gap hours like the BNetzA target scenario (0.01 h).
- **Security of supply:** with flexibility and interconnectors as in the TYNDP, the model matches the TYNDP values (10 instead of 5 h in 2030; 13 instead of 47 h in 2040). Without these levers (game rules) the same fleet has 400–550 h – which is why there are the flex card and the card for more interconnector capacity.
- **Official path (NEP B), CO_2-free share:** 97–100 % (2037) and 100 % (2045) if H_2 comes from abroad (in the game: card “Hydrogen grid & imports”).
- **Costs:** game 55–70 €/MWh for good fleets; new wind/PV plants 41–92 €/MWh (Fraunhofer ISE 2024). Today's average generation costs roughly ~100 €/MWh (own estimate: old EEG tariffs, fuel/CO_2). Total sums in the game (~€1 trillion 2026–2045) are far below whole-system studies, because grids (power grid expansion alone ~€651 bn until 2045, IMK/Hans Böckler Foundation 2024), capital of existing plants, taxes and heat/H_2 infrastructure are missing.

9. Simplifications and limits

- One grid node: no grid bottlenecks within Germany, no grid costs, no redispatch.
- Perfect foresight, no power plant outages, one weather year per stage → gaps are rather a lower bound.
- No switching of individual units (coal as one block with 15 % minimum load and 40 %/h ramp – rough, rather slightly inflexible).
- Neighbouring countries as one synthetic market; trade volumes calibrated, but no European model.
- Hydrogen: own electrolysis and caverns or imports via influence card (fixed capacity and price); no industrial H_2 demand; converted turbines burn hydrogen only (no blending).
- Flexible demand only via token and smaller than in studies; home storage not modelled.
- Biomass can shift its annual energy freely; existing gas plants do not age; one PV profile (ground-mounted ×1.10).
- No nuclear power (exit 2023), no CCS, no reservoir hydropower (0.8 GW) in Germany.
- Costs in real prices, fixed interest rate 5 %; no inflation, no distribution questions (who pays).

10. Questions and answers

Why does imported power not count as CO_2-free, even if it comes from wind?
That is how the official statistics calculate: renewable share = green power generated in Germany ÷ demand. Exported green power counts, imported power does not. We still show the CO_2 of the imports.

Why two conditions for climate protection – and why does the worse value count?

The share (EEG logic) can be gamed with exports: bots reached “100 % CO_2-free” and still emitted 40 Mt. Emissions in tonnes (Climate Protection Act) can be gamed with imports. With “the worse one counts”, neither exports nor imports help – as with net zero in reality.

Is 8 Mt in 2045 even achievable?

Hardly with own electrolysis alone (the hydrogen is not enough for the dark doldrums, old gas plants keep running). With the card “Hydrogen grid & imports”, converted gas turbines and retired old gas plants, yes – but more expensive. That is the real trade-off between climate and costs.

What does net zero mean?

Climate Protection Act: by 2045 a balance between emissions and removals by sinks (forests, moors, technology, ~40 Mt). The sinks are needed for agriculture and industry – practically nothing is left for electricity. Tonnes are counted, not shares.

Is the security of supply realistic?

The threshold (3 h) matches the German standard (2.77 h). The model is rather optimistic (perfect foresight, no outages). Without flexible demand and with few interconnectors, however, the system is harder to secure than in the studies – that is what the influence cards are for.

Why is hydrogen so cumbersome?

Round-trip efficiency ~25 %: only a quarter of the electricity comes back – hence its role as a long-term reserve. Three building blocks must fit together (electrolysis, storage, converted turbines); the storage is cheap per TWh, the electrolysis expensive. Imports via the core grid are more convenient, but expensive and only from 2035.

Are the costs realistic? My electricity price is much higher.

The game only counts generation and storage (55–70 €/MWh). Grids, taxes, levies and old support commitments are missing – together more than half of the retail price.

Why do the cost thresholds rise?

Investments are spread over their lifetime and burden all later stages; in addition the CO_2 price rises and more storage and reserve are needed.

Why more build tokens at the beginning?

The real investment need is largest at the beginning (own estimate for the EEG path: ≈ €204 bn until 2030, €74 bn 2041–45), and what is built early helps in all stages.

Why does coal always have to run?

The kept coal fleet runs at min. 15 % (district heating, mining logistics) and changes its output by at most 40 % per hour. Individual units would be more flexible, but could only restart with start-up times of several hours – the rule reflects this roughly.

Why do my old wind turbines disappear?

Plants have a lifetime (onshore wind ~25 years). By 2045 only ~30 % of today's onshore fleet is still running – replacing it is part of the task.

Where do the weather years come from?

From the European climate database PECD: 2008 is mild, 1995 low in wind, 2009 had an extreme winter (every 10–20 years). All teams get the same year.

Why is there no nuclear power or CCS?

Nuclear: legal exit in 2023. CCS: since the amendment of the Carbon Dioxide Storage Act (Bundestag, November 2025), carbon capture at gas power plants is legally possible, at coal plants it is banned. In the scenarios used here (NEP, TYNDP), CCS plays no role in the power sector; the game focuses on their options.

How accurate is the model?

Very detailed for a learning game (real weather and load data, hourly, a real optimisation model) and checked against official scenarios. But it is not a forecasting model: one grid node, simplified neighbouring countries, perfect foresight.

11. Technology and security

- Python application (NiceGUI web server) on the game master's laptop; teams only need a browser (no installation, no internet connection for the pages, just a session cookie).
- Language: German or English, chosen per browser with the **EN / DE** button at the bottom left; teams in both languages can play in the same game.
- Game master page (`/host`) only reachable on the laptop itself (otherwise `-HostRemote` + own password); default password `energiewende`.
- Team names are cleaned and shown as text; no uploads, no file sharing, no commands.
- Installation with uv: exactly the versions from `uv.lock` with checksums, only pre-built packages, waiting period for new versions; own Python, no administrator rights.
- Connection in the Wi-Fi unencrypted (HTTP) → use your own hotspot/router network, no real passwords as team names. Network problems: see "Network tips" in the README.
- Computing resources: per team run 10–20 s and up to ~1.2 GB RAM; parallel processes chosen automatically by cores and free memory (16 GB with few open programs: 2–3 processes; 5 teams ≈ 1 minute per stage).
- Windows on ARM (Snapdragon): runs via x64 emulation; Polars automatically uses the “compat” variant there.